

SURAJ BITLA

SOFTWARE ENGINEER

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SUMMARY

- Software Engineer with **5+ years of experience** in full-stack development, backend engineering, AI integration, and cloud-based deployment.
- Skilled in building scalable, production-ready systems using **React 19, Node.js, Express.js, and MySQL**, with strong focus on performance and maintainability.
- Experienced in integrating **LLM-driven features** using LangChain, GPT-4, embeddings, FAISS, and retrieval pipelines for automation and intelligent decision support.
- Hands-on with cloud infrastructure using **AWS, DigitalOcean, Ubuntu/Linux, Docker, Nginx, PM2**, and automated CI/CD pipelines.
- Background in **edge AI and cooperative inference**, optimizing deep-learning workloads on Jetson devices for real-time robotics and precision agriculture.
- Strong expertise in designing **APIs**, optimizing SQL queries, building authentication and payment flows, and deploying secure production environments.
- Proven ability to lead small engineering teams, manage project cycles, and deliver measurable improvements in reliability, onboarding, and user experience.
- Known for clear communication, ownership, and delivering complete, production-ready solutions.

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, Java, SQL, C++
Frontend:	React 19, React.js, HTML, CSS, Tailwind
Backend:	Node.js, Express.js, FastAPI, Spring Boot, REST APIs
Databases:	MySQL, PostgreSQL, Oracle DB, Redis
AI & ML:	LangChain, GPT-4, Gemini, Llama/Ollama, FAISS, sentence-transformers, RAG pipelines, PyTorch, TensorFlow
Cloud & DevOps:	AWS (EC2, S3, RDS), DigitalOcean, Docker, Nginx, PM2, GitHub Actions, Jenkins, Ubuntu/Linux
Tools:	Git, GitHub, Jira, Postman, SendGrid, Stripe, ROS, NVIDIA Jetson
Core Concepts:	Full-stack development, distributed systems, SQL optimization, CI/CD, authentication, cloud deployment, edge AI

PROFESSIONAL EXPERIENCE

Software Engineer

Kruse Construction Co. Inc., New Jersey

Feb 2025 – Present

- Designed and engineered Trakvu, a full AI-powered SaaS platform built with React 19, TypeScript, Node.js (Express), and MySQL, used by real customers for proposal comparison, contractor evaluation, and risk analysis.
- Developed complete end-to-end architecture, including frontend interfaces, backend services, database schema, authentication flows, subscription logic, and internal admin dashboards.
- Built intelligent document-processing features using LangChain, FAISS, and sentence-transformers, enabling semantic proposal comparison, risk tagging, and automated insights on contractor proposals.
- Integrated multiple LLMs (GPT-4, Gemini, Llama/Ollama) with a custom multi-model fallback system to provide reliable, cost-optimized inference in production.
- Architected and deployed a secure cloud environment with Nginx reverse proxy, PM2 clustering, SSL/HTTPS, DigitalOcean Spaces, and hardened Ubuntu configurations for stability and performance.
- Implemented Stripe subscription billing, tax logic, coupon handling, and webhook-driven invoice and onboarding email flows using SendGrid, ensuring smooth customer activation.
- Led and mentored a 3-member engineering team, improving internal workflows through code reviews, task planning, sprint execution, and development standards.
- Achieved measurable impact including an 80% reduction in manual proposal analysis effort, faster onboarding, more accurate evaluations, and increased contractor transparency.
- Built and managed additional production sites and internal tools for Kruse.

Graduate Research Assistant

Rowan University, New Jersey

Jun 2023 – Dec 2024

- Conducted research in edge AI, cooperative inference, and agricultural robotics, focusing on running deep learning models efficiently on low-power IoT and edge devices.
- Built and evaluated CV models (classification, detection, segmentation) using PyTorch and TensorFlow for precision agriculture.
- Designed split-inference pipelines that cut latency by 40% and energy by 35% across multiple model architectures.
- Developed ROS modules for NVIDIA Jetson TX2/Orin for real-time sensing, preprocessing, and inference.
- Ran large-scale benchmarks under varying compute/bandwidth conditions and produced reproducible dashboards.
- Published peer-reviewed work at IEEE ICPEC 2024 and ICEP 2025, demonstrating scalable and power-efficient agricultural AI pipelines.
- Maintained version-controlled research workflows and guided junior members.

Full Stack Engineer

Creditsafe, India

Jul 2022 – Dec 2022

- Built enterprise analytics dashboards using ReactJS, ExpressJS, Python, and Oracle DB to provide financial risk and reporting insights.
- Implemented dynamic UI rendering with pagination, lazy loading, and data-driven components, improving page performance by 40%.

- Designed and secured REST APIs for authentication, analytics, and synchronization across distributed systems.
- Automated recurring ETL processes using Python scripts, reducing manual workload by 50% and improving data consistency.
- Tuned database performance across Oracle and PostgreSQL through indexing strategies and query optimization.
- Collaborated with QA and DevOps teams in Agile sprints to ensure reliable deployments, production stability, and on-time delivery.
- Used AWS and Azure services for hosting, monitoring, and scaling microservices in production environments.

Application Development Associate

Accenture, India

Oct 2020 – Jul 2022

- Modernized legacy enterprise applications by migrating front-end modules from ASP.NET to React.js, improving maintainability and user experience.
- Developed reusable UI components, integrated APIs, and optimized SQL Server queries for multi-client enterprise systems.
- Improved CI/CD reliability by enhancing Azure DevOps pipelines, automating builds, and refining Git branching strategies.
- Participated in sprint planning, requirement analysis, architecture discussions, and code reviews within Agile teams.
- Performed functional and regression testing and resolved production issues across multiple applications and client accounts.
- Delivered features across 10+ sprints and mentored junior associates through knowledge-transfer sessions on React, version control, and deployment workflows.

Full Stack Developer

Cipla, India

Oct 2019 – Oct 2020

- Developed and maintained internal enterprise applications using Java, Spring Boot, and a microservices architecture to support manufacturing, compliance, and operations workflows.
- Created secure backend APIs integrated with ERP/MIS systems for real-time tracking of manufacturing and operational data.
- Improved PostgreSQL performance by optimizing queries, adding indexes, and restructuring slow-running analytical operations.
- Built dashboards using React.js to visualize plant metrics, reports, and production KPIs for internal teams.
- Managed deployments using Jenkins and Docker on AWS EC2/RDS, ensuring stable and reliable production environments.
- Worked closely with QA and DevOps teams to refine deployment pipelines, reduce release issues, and maintain high uptime.

EDUCATION

Rowan University, NJ, USA

Master of Science in Computer Science 4.0/4.0

Jan 2023 – Dec 2024

Sreenidhi Institute of Science & Technology, India

Bachelor of Computer Science and Engineering 9.3/10

Aug 2016 – Jun 2020

PROJECTS

Forgetting LLM System — Machine Unlearning Framework ([Link](#))

- Built a machine-unlearning system removing sensitive data from LLMs quickly without full retraining using modular workflows.
- Created flexible forgetting tools for pattern removal, user deletion, and retention rules supporting fast scalable ops.
- Developed React interface and Docker deployment with CI/CD ensuring fast inference, stable tests, and monitoring efficiency.

Real Estate Sales Analysis ([Link](#))

- Built a Databricks pipeline to clean, transform, and aggregate Connecticut real-estate sales data at scale.
- Analyzed trends like pricing, property types, sales ratios, and high-activity regions using SQL and Python.
- Delivered forecasts and insights through visual dashboards, using AWS S3 and EC2 for efficient scalable ETL.

Cooperative Inference Framework (SplitTracr & Precision Agriculture) ([Link](#))

- Built a cooperative edge-server inference framework for precision agriculture to split deep learning workloads efficiently.
- Partitioned models like AlexNet, ResNet, Faster R-CNN, and Mask R-CNN for real-time performance with reduced load.
- Designed adaptive offloading strategies reacting to network, compute, and latency conditions to improve speed and energy use.

E-Commerce Platform – ShopRight ([Link](#))

- Developed a scalable e-commerce platform with secure user authentication, role-based authorization, product catalogs, cart management, and order tracking.
- Implemented JWT-based authentication and session handling to protect user data and support multi-role access (customers, admins).
- Designed a responsive frontend with ReactJS focused on fast product search, category filters, and smooth checkout experiences.
- Built backend APIs with Node.js and Express, using optimized MySQL queries to support 1K+ concurrent users in peak conditions.
- Deployed services on AWS using containerized workloads and CI/CD pipelines, reducing release cycle time by 40% and improving reliability.

CERTIFICATIONS

- [Microsoft certified Azure fundamentals](#)
- [Microsoft Certified Power Platform Fundamentals](#)
- [Machine learning with Python - IBM](#)
- [Data Science Methodology – IBM](#)
- [AWS Cloud technical essentials](#)
- [Python for Data Science & AI Development - IBM](#)
- [Machine learning & Data Science with Python - Udemey](#)
- [Migrating to AWS Cloud](#)